

Harshil Agrawal

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EDUCATION

Virginia Tech | M.S. Data Science | Blacksburg, VA | Aug 2026 to May 2028

SVKM's NMIMS University, MPSTME, Mumbai | B.Tech Computer Engineering, GPA: 3.60/4.0 | Graduated May 2026

EXPERIENCE

Prowess Consulting | Data & AI Engineering Intern | Dec 2025 - Jun 2026

- Owned end-to-end delivery of an LLM-powered WhatsApp ordering platform on Azure App Service for pharmaceutical distribution.
- Integrated Azure OpenAI intent classification with a 1,500+ product catalog and Snowflake-backed conversation logging.
- Built the SalesVitals FastAPI analytics stack with SQLite snapshot mode, eliminating recurring Snowflake costs.
- Maintained reproducibility with Git, GitHub Actions, and automated test suites.

Gram Panchayat, Maharashtra | Social Impact Data Analyst, Volunteer | 2024

- Collected and analyzed community participation, sanitation, and education data for rural development.
- Tracked 500+ sapling plantations and evaluated sanitation and digital-literacy programs.
- Worked with local authorities on resource allocation for old-age homes and orphanages.

SELECTED PROJECTS

Banking PII Protection | github.com/Harshil411/banking_PII

- Built a local-first FastAPI and React service for PII extraction, schema validation, and anonymized output. Reported micro-F1: 0.866 across 32,017 labeled entities.

Manufacturing Defect Prediction | github.com/Harshil411/portfolio

- Deployed an inspectable Random Forest classifier through FastAPI. The project uses synthetic manufacturing sensor data and documents its model behavior and evaluation.

RESEARCH

A Hybrid LSTM + Deep Reinforcement Learning Framework for Enterprise-Grade Decision Support and Automated Trading

IEEE ICCICT 2026, under publication

- Proposes a hybrid LSTM and Deep RL framework that separates forecasting from execution.
- Validated on NIFTY-50 backtests with a reported Sharpe ratio of about 1.8 using LSTM, PPO, DQN, and SAC.

PROFESSIONAL DEVELOPMENT AND AWARD

Certificate of Participation in ACM SemiCode | Unstop | Mar 2024 | Credential ID: da0389db-6732-4f51-a3fb-4b97ec5dfe93

Workshops: Power BI Workshop; AI Prompting and N8N Automation Workshop, Prowess Consulting; AI and HPC Workshop, KLA and IIT Madras.

Award: Best Executive in Public Relations, Google Developer Student Clubs, NMIMS.

SKILLS

Programming: Python, SQL, Java, C, C++, JavaScript | Backend: FastAPI, Spring Boot, Spring Cloud, REST API | Cloud: Azure, Docker, GitHub Actions, CI/CD | Data and ML: Snowflake, SQLite, LSTM, Deep RL, scikit-learn, pandas